

Problem Statement

For $a \in \mathbb{R}$, $|a| > 1$, let

$$\lim_{n \rightarrow \infty} \left(\frac{\sum_{r=1}^n \sqrt[3]{r}}{n^{7/3} \sum_{r=1}^n \frac{1}{(an+r)^2}} \right) = 54.$$

Find the sum of squares of all possible values of a .

Detailed Solution

This limit can be evaluated by transforming both the numerator and the denominator into Riemann sums, which can then be expressed as definite integrals. Let's break this down step-by-step.

Step 1: Algebraic manipulation of the limit

Let L be the given limit:

$$L = \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r^{1/3}}{n^{7/3} \sum_{r=1}^n \frac{1}{(an+r)^2}}$$

We want to manipulate the expressions into the form $\frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right)$, which is the standard definition of a Riemann sum that converges to $\int_0^1 f(x) dx$.

First, factor out n^2 from the denominator of the summand in the bottom expression:

$$\frac{1}{(an+r)^2} = \frac{1}{n^2 \left(a + \frac{r}{n}\right)^2}$$

Substituting this back into the limit yields:

$$L = \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r^{1/3}}{n^{7/3} \sum_{r=1}^n \frac{1}{n^2 \left(a + \frac{r}{n}\right)^2}} = \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r^{1/3}}{n^{1/3} \sum_{r=1}^n \frac{1}{\left(a + \frac{r}{n}\right)^2}}$$

Now, we multiply the numerator and the denominator by $\frac{1}{n^{4/3}}$ to perfectly set up both Riemann sums:

$$L = \lim_{n \rightarrow \infty} \frac{\frac{1}{n^{4/3}} \sum_{r=1}^n r^{1/3}}{\frac{1}{n^{4/3}} n^{1/3} \sum_{r=1}^n \frac{1}{\left(a + \frac{r}{n}\right)^2}} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^{1/3}}{\frac{1}{n} \sum_{r=1}^n \frac{1}{\left(a + \frac{r}{n}\right)^2}}$$

We can now split the limit into the quotient of two separate limits:

$$L = \frac{\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^{1/3}}{\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{1}{\left(a + \frac{r}{n}\right)^2}}$$

Step 2: Evaluate the Riemann sums as integrals

We recognize both limits as Riemann sums over the interval $[0, 1]$.

For the numerator:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^{1/3} = \int_0^1 x^{1/3} dx$$

Evaluating this integral gives:

$$\int_0^1 x^{1/3} dx = \left[\frac{3}{4} x^{4/3} \right]_0^1 = \frac{3}{4}$$

For the denominator:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \frac{1}{\left(a + \frac{r}{n}\right)^2} = \int_0^1 \frac{1}{(a+x)^2} dx$$

Evaluating this integral gives:

$$\int_0^1 (a+x)^{-2} dx = \left[-\frac{1}{a+x} \right]_0^1 = -\frac{1}{a+1} - \left(-\frac{1}{a+0} \right) = \frac{1}{a} - \frac{1}{a+1}$$

Finding a common denominator yields:

$$\frac{1}{a} - \frac{1}{a+1} = \frac{(a+1) - a}{a(a+1)} = \frac{1}{a(a+1)}$$

Step 3: Solve for a

Substitute the values of the evaluated integrals back into the expression for L :

$$L = \frac{\frac{3}{4}}{\frac{1}{a(a+1)}} = \frac{3}{4} a(a+1)$$

We are given that $L = 54$. Setting our expression equal to 54 gives:

$$\frac{3}{4} a(a+1) = 54$$

Multiply both sides by $\frac{4}{3}$:

$$\begin{aligned} a(a+1) &= 54 \cdot \frac{4}{3} \\ a^2 + a &= 72 \end{aligned}$$

Bring all terms to one side to form a standard quadratic equation:

$$a^2 + a - 72 = 0$$

Factoring the quadratic equation gives:

$$(a+9)(a-8) = 0$$

So, the possible values for a are $a = -9$ and $a = 8$. We must check if these satisfy the initial condition $|a| > 1$:

- For $a = -9$, $|-9| = 9 > 1$ (Satisfied)
- For $a = 8$, $|8| = 8 > 1$ (Satisfied)

Step 4: Compute the final answer

The problem asks for the sum of the squares of all possible values of a .

$$(-9)^2 + (8)^2 = 81 + 64 = 145$$

Final Answer:

145